

6 Agentic AI Workflow Orchestrator (n8n + OpenAI)

Title: Autonomous Agentic Pipelines with Self-Correction & Multi-Step Reasoning

Role: AI Engineer / Orchestrator

Tech Stack: n8n, OpenAI (GPT-4), LangChain, Python, JSON Schema

Timeline: 2025 – Present

Executive Summary

"Moving beyond chatbots to digital employees."

Most AI implementations today are passive—they wait for a user prompt. **Agentic AI** is active; it receives a high-level goal (e.g., "Research this company and draft a personalized outreach") and autonomously figures out the steps to achieve it.

This project involved architecting robust, self-healing workflows using **n8n** as the orchestration backbone and **LLMs** as the reasoning engine. The system can browse the web, scrape data, make API calls, and—crucially—**fix its own mistakes** if a step fails.

Core Technical Challenge: "The Hallucination of Action"

When LLMs are asked to execute code or call APIs, they often hallucinate parameters or generate invalid JSON.

- **My Solution:** I implemented a **"Loop-on-Error" architecture**. If the AI generates an output that causes a downstream error (e.g., invalid JSON format), the error message is fed *back* into the AI with the prompt: *"You caused Error X. Fix your syntax and try again."*

System Architecture

The architecture mimics a human cognitive loop: **Plan → Execute → Evaluate → Refine**.

1. The Planning Layer (The Brain)

- **Model:** GPT-4o / Claude 3.5 Sonnet.
- **Function:** Receives a vague user request and decomposes it into a **Structured DAG (Directed Acyclic Graph)** of tasks.
- **Technique:** Used **Chain-of-Thought (CoT)** prompting to force the model to output its

reasoning strategy before generating the action payload.

2. The Execution Layer (The Body - n8n)

- **Tool Use:** The AI is given access to specific "Tools" via n8n nodes:
 - **Web Scraper:** To gather live context.
 - **HTTP Request:** To interact with external CRMs or databases.
 - **Data Transformer:** To normalize unstructured text into structured JSON.
- **Security:** Implemented "**Human-on-the-Loop**" checkpoints for high-risk actions (e.g., sending emails), where the agent drafts the action but waits for approval.

3. The Reliability Layer (Self-Correction)

- **JSON Validation:** Enforced strict **JSON Schemas** (Pydantic/Zod logic) on model outputs.
- **Error Handling:** Standard automation stops on error. This system treats errors as new inputs.
 - *Scenario:* Agent tries to scrape a 404 URL.
 - *Response:* System catches 404 → Feeds back to Agent → Agent generates a Google Search query to find the *correct* URL → Retries scraping.

4. Memory & Context

- **Short-Term Memory:** Maintains a conversation buffer so the agent remembers previous steps in the workflow.
- **Long-Term Memory:** Connected to a **Vector Store (Pinecone)** to retrieve past successful strategies for similar tasks.



Tech Stack Details

- **Orchestration:** n8n (Self-Hosted for data privacy).
- **LLM Ops:** OpenAI API (Function Calling mode).
- **Scripting:** Python nodes within n8n for complex data transformation.
- **Data Interchange:** Strict JSON formatting.



Strategic Impact

1. **Operational Velocity:** Reduced multi-hour manual research processes to minutes of autonomous execution.
2. **Robustness:** Proved that non-deterministic AI can run reliably in deterministic enterprise environments if wrapped in **self-correction loops**.
3. **Future-Proofing:** Positioned yourself at the forefront of the "**Agentic Web**"—the next evolution of the internet where agents, not humans, consume interfaces.